

Contact

322 W 4th St, Cincinnati, OH - 45202
219-809-1518 (Mobile)
rudrakshsy@gmail.com

www.linkedin.com/in/rudraksh-syal
(LinkedIn)
github.com/rudrakshsy (Portfolio)

Top Skills

Pricing Optimization
Demand Forecasting
Time Series Forecasting

Languages

English (Full Professional)
Hindi (Native or Bilingual)
Punjabi (Native or Bilingual)

Certifications

Machine Learning A-Z™: Hands-On
Python & R In Data Science

Patents

Efficient resource scheduling using
adaptive scheduling criteria

Rudraksh Syal

Senior Data Scientist | Product Analytics, Experimentation & Applied
ML | Pricing, Forecasting, Causal Inference & GenAI
San Jose, California, United States

Summary

Senior Data Scientist with 10 years of experience building ML systems at the intersection of pricing, forecasting, and applied AI. For the past 4.5 years at Vista Tech (Vista Global, \$2.4B revenue), I've served as the DS owner of fleet-wide pricing and demand forecasting across a 2,100+ aircraft global network. My work spans the full ML lifecycle — from economic problem formulation and model development to production deployment and ongoing MLOps. Recent highlights include architecting a supply-demand equilibrium Dynamic Pricing Engine with 10+ interconnected forecasting models, and shipping two production LLM pipelines — one for parsing 1,100+ program member contracts, another powering live cancellation policy badges on the Vista website. Over time my role has expanded beyond pure data science into ML engineering and data infrastructure — setting up AWS MWAA Airflow, building containerized SageMaker pipelines, and owning end-to-end automated workflows for my team. Before Vista, I spent two years at 84.51° (Kroger's data science subsidiary) applying Bayesian causal inference and combinatorial optimization to CPG merchandising decisions at national retail scale. I hold a co-inventor credit on US Patent No. 12,314,757 B1 (adaptive resource scheduling, granted 2025), an MS in Business Analytics from Purdue University, and a dual degree in Mathematics from BITS Pilani. I founded a Knowledge Café (peer learning program) delivering 8 cross-functional sessions to 60+ colleagues across product, engineering, and ops, and have mentored junior data scientists for the last 2 years through building production systems.

Experience

Vista

4 years 10 months

Senior Data Scientist (Pricing Data Science)

January 2023 - July 2026 (3 years 7 months)

San Jose, CA

Dynamic Pricing Engine | Python, Snowflake, AWS SageMaker, Airflow, Docker

- Architected and deployed a supply-demand equilibrium pricing system for Vista's US on-demand private jet fleet. The engine solves for the optimal live yield per departure date by equating forecasted on-demand supply against price-elastic demand, grounded in an empirically estimated price elasticity.
- Built 10+ interconnected sub-models from scratch: XGBoost program booking forecast (MAE ~3.5 live legs, MAPE ~20%); autoregressive model for unscheduled maintenance (UMX); moving average models for AOG, SMX, and CNA unavailability.
- Led the full project lifecycle solo — economic problem formulation through MVP, multi-fleet scaling, and production deployment — with one junior DS collaborator; system is actively used by the US fleet pricing team.

LLM Pipelines — Contract Parsing & Cancellation Policy Extraction | Python, Claude API, Snowflake, Airflow

- Built a production LLM pipeline to parse 1,100+ program member contracts (20–30 pages each), extract structured term categories, and load results into Snowflake tables subsequently used by engineering to build customer-facing online modules for sales reps.
- Separately built a cancellation policy extraction pipeline processing ~600 operator policy blurbs, delivering structured output and statistics to Snowflake; results now surfaced as live human-readable cancellation badges on each aircraft listing on the XO marketplace.

Leadership & Knowledge Sharing

- Founded and ran the Knowledge Café initiative — 8 cross-functional sessions reaching 60+ colleagues across product, engineering, ops, and leadership — covering MLOps, SageMaker AutoML, model monitoring, and deployment best practices.
- Mentored junior talent over 2+ years across multiple projects including the Member Program Forecast pipeline and the Dynamic Pricing engine.

Senior Data Scientist (Pricing Intelligence)

October 2021 - December 2022 (1 year 3 months)

California, United States

Price Elasticity in Private Jet Bookings | Python, AWS SageMaker

- Estimating price elasticity as the change in booking conversion rate wrt change in trip pricing

- Developed an RF model to predict the relationship of Conversion% with trip attributes such as Quotes, Callout, EFT, and Yield, with a 79% test accuracy
- Implemented Dynamic Pricing Engine with conversion function as the input & generating better contribution v/s holding a constant price in the past

Booking Propensity Model via High-Intent User Analysis | Python, AWS, Salesforce

- Evaluated the willingness of a lead to book based on their consumer behavior in the past 3/7/10-day activity period (Time to Conversion analysis)
- Trained an XGBoost classification model with 0.78 AUC to categorize leads into Hot/Warm/Cold, based on their predicted probabilities
- Pushed the lead scores to Salesforce, set up SFDC Campaigns to track a ~1.2% conversion rates for 80%+ predicted probabilities, a.k.a. Hot Leads

--> Metric Definition and Analysis Expertise: Spearheaded 'session duration' metric definition and implementation, enhancing user engagement insights through advanced analysis of complex interactions

--> Technical Proficiency in Event Tracking with Mixpanel: Led Mixpanel integration for in-depth event tracking, providing key insights for app optimization and user experience improvements

--> Hackathon Success in Document Analysis: Led a hackathon project using LLM models for actionable insights from PDF contracts, securing 2nd place to develop a novel tabular output for data interpretation

84.51°

2 years 4 months

Senior Data Scientist (Space Planning & Optimization)

December 2020 - October 2021 (11 months)

Cincinnati, Ohio, United States

Bakery Space Optimization | Python, Databricks, Azure

- Helped Kroger optimize the relative space allocated to 4 Bakery commodities (Bread, Breakfast Sweets, Cakes, & Cookies) for a new Kroger store, thereby maximizing the bakery sales for a combination of these 4 commodities
- Developed independent sales prediction models for each commodity with 26 store-level variables
- Utilized Integer Optimization to get the precise allocation of space for the 4 commodities

Front-end Battery Destination Analysis | Python, Databricks, Kroger PLM Data

- Led an end-to-end analysis to understand the battery footprint at Kroger stores and eventually reduce the size of check-lane battery endcaps
- Identified multiple segment of stores w/ battery destination in different locations and guide the impact of space taken from battery endcaps
- Impact measured in terms of store-level KPIs such as \$ Sales, Visits, Units, Households, and so on

Senior Data Scientist (Merchandizing Analytics)

July 2019 - October 2021 (2 years 4 months)

Cincinnati, Ohio, United States

Causal Inference Measurement for Quaker Endcap | R, Tableau, Causal Impact

- Measure the incremental effect of in-store tests using structural time-series models (BSTS)
- Impact measured by the difference in performance of test v/s synthetic control
- For 3 Quaker breakfast options, there was a mere 4% uplift in sales as compared to 10% uplift in Endcap sales, leading to insignificant results

Adjacency Optimization for Flavored Spirits | Python, Spark, GCP

- Design optimal shelf flow by analyzing customer cross-shop at household/ basket level
- Quantify how customers shop spirits via association scores and build an optimization model for (spirits') flavors merchandizing
- Recommended an assortment plan for spirit aisles - to keep/not keep Flavored Spirits together

Frito Lay Baked brand Profiling & Migration Analysis | SQL, R, Tableau

- Exploratory analysis to contrast the 'Premium' Snack consumers to the Baked shoppers
- Profiled/Indexed the prevalence of 'FDA Reduced Fat Package Claims' for 4 Baked sub-lines
- Aided Pepsi to realize the significance of introducing 'Baked' brand v/s the 'Mainstream' options in their quest for a "healthier" competitor of Pringles

Corteva Agriscience

Data Scientist

December 2018 - April 2019 (5 months)

Greater Indianapolis

- Developed an intelligent Agribusiness Recommendation Optimization Engine: AGRO, to identify potential customers, understand market needs, and provide customized & accurate marketing recommendations regarding individual customers (farmers).
- Utilizing a blend of multiple machine learning algorithms, this cross-validated engine ensures generated recommendations are optimized, inaccurate suggestions are minimized, and precisely predict which segment of customers to focus with what category of products.
- Achieved an accuracy of 82% with Logistic Regression algorithm thereby focusing primarily on increasing the Sensitivity of the model as we did not want to lose any potential leads. Furthermore, attained an accuracy of 75% in predicting the type of Pesticide (say Herbicide, Nematicide, etc.) a farmer will buy in future.

Practo

2 years 11 months

Data Scientist (Product Data Science)

January 2017 - April 2018 (1 year 4 months)

Bengaluru Area, India

- R / Shiny Application Development for Product assessment | R, ShinyApps
 - Developed a real-time web Shiny application in RStudio incorporating all significant metrics of the 'Search' platform, replacing a \$4000 per month per person third-party SQL software, Periscope
- Feedback 'Most Helpful Reviews' Algorithm | R, Google NLP API – Advancing towards a mature algorithm for 'Most Helpful Reviews' across the platforms, to achieve optimization between the consumer and supply side needs. (Doctors & Patients)
- Product Conversion Strategy – Developing strategies for the growth in Conversion% across the platforms, as well as drawing insights from the Tier-1/ Tier-2 markets for consumer behaviour on the website.
- Customer Funnel & Conversion% Optimization via Experimentation | A/B Testing, R
 - Evaluating & fixing drops in conversion% in customer journeys via Hypothesis Testing

- Choosing accurate KPIs, designing, running & analyzing experiments for online feature changes

Data Scientist (Sales Business Intelligence)

June 2015 - December 2016 (1 year 7 months)

Bengaluru Area, India

- Doctor Interactions' Sentiment Analysis | R, NLP - Built a framework for Text Mining in R for the Sales Interactions & Product Feedback data involving Sentiment Analysis and behavior analytics to set up ground rules (whether customer usage is positive/negative)
- Lead Segmentation Scoring via Predictive Analytics | R, Regression – Built a mathematical model in R to determine buying probability score for all leads. Devised a geography specific categorization algorithm to create focused lead lists to publish scores in Salesforce CRM leading to a 30% increment in sales
- Dashboard Analytics – Prepared real time Interactive Sales & Product Dashboards to track performance, effort and compliance related field data to plan effective interventions and achieve the maximum efficiency.

Practo

Business Analyst Intern

July 2014 - December 2014 (6 months)

Bengaluru Area, India

- Incentive Modelling – Streamlined the 'Revenue v/s Incentive Model' and automated the process of Incentive pay-out.
- Market Research – Researched on Practo's major competitors and carrying out a Competitor Analysis to list out the USPs of our product.

Essar Power

Summer Research Intern

May 2013 - July 2013 (3 months)

India

- ABT Reporting – Completed a project on 'Availability Based Tariff'(ABT) and 'Regulation of UI Charges' under the Finance Department of the company.

Education

Purdue University

Master's degree, Business Analytics and Information Management (BAIM) -
Data Science Track · (2018 - 2019)

Birla Institute of Technology and Science, Pilani
Master's degree, Applied Mathematics · (2011 - 2015)

Saint Xavier's Senior Secondary School
High School, Economics, Mathematics and Science · (1999 - 2011)